Samuel Khosla

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Education

University of Toronto — Toronto, ON
Honours Bachelor of Science in Mathematics — *Expected graduation: May 2030*

Work Experience

Ski Lift Operator — Jan. 2024 – Mar. 2024
Camp Fortune — *Chelsea, QC*

- Ensured the safe usage of ski lifts by Camp Fortune customers by taking quick action to prevent accidents
- Reduced ski lift wait times by assisting customers with technical difficulties in a patient and friendly manner

Projects

Blackjack Basic Strategy Calculator | *Python, Flask, HTML, CSS, Javascript* — Aug. 2026 – Present

- Built a full-stack blackjack basic strategy solver that computes optimal-EV decisions via dynamic programming
- Implemented a composition-independent EV engine in Python, weight-averaging outcomes across thousands of two-card hand combinations per dealer upcard to match real-world strategy charts
- Designed a Flask REST API serving rule-configurable strategy computations as JSON to a dynamic frontend
- Built a JS/HTML/CSS frontend rendering live strategy tables and EV calculations with debounced API calls

Word Hunt Solver | *HTML, CSS, Javascript* — Jul. 2026 – Present

- Built a Word Hunt/Boggle solver using trie-based DFS to find all valid words on 4×4–6×6 boards
- Implemented dynamic SVG path visualization showing each word's letter path on the board on hover
- Designed a JS/HTML/CSS UI supporting custom dictionary uploads, pasted word lists, and a default dictionary

Awards & Honours

Canadian Mathematical Olympiad (CMO) Qualifier — Feb. 2026
Canadian Mathematical Society

- One of 95 students (1.8% of Canadian Open Mathematics Challenge contestants) who qualified from the COMC

American Invitational Mathematics Examination (AIME) Qualifier — Dec. 2026
Mathematical Association of America

- Qualified from a high score in the American Mathematics Competition (AMC 12)

Canadian Senior Mathematics Contest: Group III — Dec. 2025
University of Waterloo

- Scored in the top 0.6% of over 12,000 participating high school students across Canada

Extracurriculars

Program in Mathematics for Young Scientists (PROMYS) Attendee — Jun. 2025 – Aug. 2025
Boston University — *Boston, MA*

- One of 80 (5% of applicants) to be selected to attend an intensive summer mathematics program
- Worked with peers to complete daily number theory problem sets at the university level
- Attended minicourses by counsellors on topics such as circle-packing, p-adic numbers, and quartic reciprocity
- Researched and delivered a group presentation on number-theoretic properties of binomial coefficients

Exploring Mathematics and Computer Science (EMACS) Camp Attendee — May 2025
University of Waterloo — *Waterloo, ON*

- Selected to attend based on top scores on the 2025 Fermat Contest and the 2024 CSMC
- Participated in group problem-solving sessions and attended math and CS lectures led by university faculty

Technical Skills

Languages: C++, Python, JavaScript, HTML/CSS, Excel/Google Sheets, LaTeX
Frameworks and Tools: Flask, Git, VS Code, PyCharm, Linux